

A Reproducible Framework for Interpreting Calibration Slopes in External Model Evaluation

Abstract Calibration slopes are frequently reduced to a binary adequacy label even though their meaning depends on the definition of the linear predictor, the evaluation sample, and the fitted recalibration model. This methodological article sets out a reproducible assessment framework that separates overall calibration, spread of predictions, local agreement, discrimination, and decision consequences. The framework requires investigators to freeze the original prediction rule, declare the target population and prediction horizon, generate predictions without refitting, estimate calibration quantities under an explicit model, and report uncertainty together with graphical assessments. A slope below one indicates predictions that are generally too extreme, whereas a slope above one indicates predictions that are insufficiently dispersed, provided that the slope is estimated on the original model's linear predictor. Recalibration, model updating, subgroup assessment, and clinical-utility analysis are treated as distinct tasks. The framework is intended to improve the traceability and interpretation of external evaluations; it does not convert statistical calibration into evidence of clinical usefulness.

Keywords calibration slope; external validation; prediction model; reproducibility; reporting

1 Introduction

External evaluation asks whether a prediction model retains adequate performance when applied to data that were not used to develop it. Calibration is central to that question because many clinical decisions depend on agreement between predicted probabilities and observed risks, not only on the ordering of individuals. Nevertheless, evaluations often report one discrimination statistic and one calibration slope while omitting the precise model version, prediction horizon, analysis set, or scale on which recalibration was performed [1,2]. These omissions make apparently comparable estimates answer different questions.

The calibration slope is particularly vulnerable to reversed interpretation. When the slope is below one, differences in the original linear predictor are generally larger than supported by the evaluation data: high predictions tend to be too high and low predictions too low. When the slope exceeds one, the predictions vary too little and are concentrated toward the average risk [2,3]. These statements assume that the original predictor has been preserved and entered into a correctly specified recalibration model. A slope fitted after arbitrary rescaling, predictor refitting, or undocumented preprocessing does not retain the same interpretation.

A calibration slope is also not a complete model evaluation. Calibration-in-the-large addresses systematic overprediction or underprediction, flexible curves describe local agreement, discrimination describes ranking, and decision-analytic measures assess consequences under specified thresholds. A model can rank individuals well while producing poorly calibrated probabilities, and a recalibrated model can remain clinically unhelpful [1,4]. Each quantity therefore needs a declared estimand, an appropriate analysis set, and uncertainty that reflects the evaluation design.

The objective of this article is to define a practical evidence chain for interpreting calibration slopes in external model evaluation. The framework links the frozen prediction specification, target setting, evaluation dataset, fitted performance objects, machine-readable results, figures, and manuscript statements. It is designed as a neutral author-manuscript framework rather than a journal-layout template and is applicable to binary and time-to-event predictions with outcome-specific adaptations.

2 Scope and estimands

2.1 Target population, setting, and prediction horizon

An external evaluation begins by defining where and for whom the model is intended to be used. Eligibility criteria, care setting, time zero, outcome definition, competing events, and prediction horizon must

be stated before performance is examined. The analysis set should be derived from these definitions rather than from complete-case convenience. If several horizons or settings are clinically relevant, each constitutes a distinct evaluation target and should not be combined into one generic calibration statement.

2.2 The prediction object

The evaluated object comprises more than a list of coefficients. It includes predictor definitions, measurement timing, transformations, coding rules, coefficients, intercept or baseline hazard, and any preprocessing learned in the development data. These components are frozen before evaluation. Predictions are generated without refitting the original coefficients. If a required transformation or baseline quantity is unavailable, the model is not fully transportable and the missing component should be reported rather than silently reconstructed from the evaluation sample.

2.3 Calibration estimands

Calibration-in-the-large compares the average predicted risk with the observed risk and captures a systematic shift. The calibration slope evaluates whether the spread of the original predictions is appropriate. A flexible calibration curve examines local agreement across the prediction range. These estimands are related but not interchangeable. For survival outcomes, their definitions must specify the horizon and a method that accounts for censoring; for competing-risk outcomes, the predicted and observed quantities must refer to the same cumulative-incidence estimand.

3 Evaluation framework

Table 1 Core components of a reproducible external calibration assessment

Component	Required specification	Primary check	Interpretive boundary
Prediction rule	Version, coefficients, intercept or baseline hazard, transformations	Reproduce predictions on test inputs	Do not refit during evaluation
Evaluation target	Population, setting, time zero, outcome, horizon	Match eligibility and endpoint definitions	Do not merge distinct targets
Analysis set	Exclusions, missing data, censoring, competing events	Account for all eligible records	Complete cases may change the target
Calibration	Intercept, slope, curve, uncertainty	Use the original prediction scale	One summary cannot describe local agreement
Clinical utility	Decision, threshold range, consequences	Compare decision strategies	Calibration alone is insufficient
Traceability	Model object, result record, table, figure, text	Reconcile values and direction	Do not type corrections into outputs

3.1 Preserve the original prediction scale

For a logistic model, the original linear predictor is commonly entered as the sole covariate in a recalibration model. Estimating an intercept with the slope fixed at one assesses calibration-in-the-large; estimating both intercept and slope assesses the spread of predictions. The exact parameterization and link function should be reported. The same principle applies to survival models, but the baseline hazard, prediction horizon, and censoring-aware estimation must remain aligned with the original risk definition.

Preprocessing estimated in the development data, including centering, scaling, category cut points, and imputation models, is part of the frozen prediction rule. Re-estimating these quantities in the external dataset can improve apparent fit while changing the object under evaluation. Such a procedure may be legitimate model updating, but it must be labeled and analyzed separately from the untouched-model assessment.

3.2 Define the analysis set and handle missing data

The report should account for the number of eligible individuals, exclusions before prediction, unavailable predictors, missing outcomes, and records included in each performance calculation. Missingness can alter both case mix and calibration. An imputation strategy should preserve the distinction between information available in the development model and information learned from the evaluation sample. Sensitivity analyses are appropriate when plausible missing-data assumptions lead to materially different performance conclusions.

For time-to-event outcomes, censoring and competing events require additional care. The observed outcome estimate used in a calibration plot must correspond to the same horizon and event definition as the predicted risk. Numbers at risk, follow-up distribution, and event counts provide essential context for uncertainty. A nominally large cohort may still yield imprecise calibration estimates if few events occur in clinically important parts of the prediction range.

3.3 Quantify uncertainty and inspect local agreement

Calibration estimates should be accompanied by confidence intervals generated by a method appropriate to the sampling and outcome structure. A point estimate that differs from the ideal value can remain compatible with substantial sampling uncertainty. Conversely, a narrow interval around a non-ideal value can identify a reproducible mismatch even when the absolute numerical departure appears modest. Interpretation should therefore combine effect size, precision, and potential decision consequences.

A calibration curve should retain the probability scale and show the region in which evaluation data are available. Smoothing method, grouping, boundary behavior, and extrapolation can alter the visual impression. The curve should be displayed with uncertainty and the distribution of predicted risks. Decorative gradients, adequacy badges, or grouped observed-to-expected ratios cannot replace a curve when the scientific question concerns local agreement [2,3].

3.4 Separate evaluation, recalibration, and updating

The untouched model should be evaluated first. Recalibration of the intercept or slope, revision of selected coefficients, addition of predictors, and full redevelopment represent progressively more extensive updating strategies. Their apparent performance should not be reported as if it described the original model. Updated models require internal validation within the updating dataset and, ideally, evaluation in further data because the updating step consumes information and can overfit.

Subgroup analyses also answer additional questions. A global calibration slope can conceal clinically important variation, but subgroup-specific estimates are often unstable and can encourage post hoc interpretation. Subgroups should be prespecified when possible, supported by adequate sample size and event counts, and accompanied by interaction or heterogeneity assessments that reflect the intended use rather than arbitrary significance thresholds.

4 Reporting and interpretation

4.1 Minimum numerical and graphical reporting

A complete report identifies the model and version, target population, setting, outcome, prediction horizon, eligible sample, analysis set, number of events, missing-data handling, and prediction distribution. It reports calibration-in-the-large, calibration slope, confidence intervals, discrimination, and a calibration curve when these quantities are relevant. For survival outcomes, the amount of follow-up and censoring should be described. Each table and figure should state the evaluated model and analysis set rather than relying on a generic caption.

The manuscript should preserve the direction of each estimate. A slope below one may be described as predictions that are too extreme on average; a slope above one may be described as predictions that are insufficiently dispersed. Neither finding identifies which predictor coefficients are wrong, proves that development overfitting was the cause, or establishes that a particular updating strategy will improve decisions. Causal language is inappropriate unless supported by a design and analysis that address a causal question.

4.2 Traceable assembly of results

Numerical results should be exported from the fitted evaluation objects to a stable machine-readable record. Tables, figures, and manuscript statements should read from that record or from verified derivatives. Corrections then return to the analysis that generated the estimate instead of being typed into separate outputs.

At minimum, validation should reconcile sample sizes, event counts, model version, estimate direction, interval limits, rounding, and labels across the result record and all public-facing products.

A reproducible record does not guarantee a scientifically valid analysis. It makes the analysis inspectable. Design choices, outcome definitions, exclusions, missing-data assumptions, and updating decisions still require domain justification. Automated checks are most useful when they protect these declared contracts and stop on mismatches, while substantive interpretation remains tied to the clinical question and the evidence available in the evaluation setting.

5 Discussion

The proposed framework treats external calibration as an evidence chain rather than a single diagnostic number. Its central requirement is to preserve the identity of the prediction model while making the target setting and performance estimands explicit. This separation prevents an updated model from being mistaken for the original model and prevents a change in case mix, horizon, or outcome definition from being described as a universal property of model quality.

The framework also clarifies why calibration slope direction must be reported carefully. A slope below one indicates excessive spread of the original predictor relative to the evaluation data, whereas a slope above one indicates insufficient spread. The labels do not supply a mechanism. Differences in case mix, measurement, predictor effects, treatment patterns, or outcome incidence may contribute. Investigation of those mechanisms requires prespecified analyses and cannot be inferred from the slope alone.

Clinical utility remains a separate endpoint. Calibration affects the credibility of predicted probabilities, but a well-calibrated model may not improve decisions if it does not change management or if the consequences of errors outweigh benefits. Conversely, a model with imperfect overall calibration might retain value after transparent recalibration in a setting where its ranking information and decision consequences have been independently assessed [1,4]. Decision-curve or other utility analyses should therefore specify the action, threshold range, and relative consequences rather than using calibration as a proxy.

Several limitations should be recognized. The framework does not prescribe one estimator for every outcome or sampling design, and additional methods are needed for clustered data, complex surveys, dynamic predictions, recurrent events, and competing risks. It also does not provide universal sample-size thresholds. Precision depends on the prediction distribution, event frequency, censoring, and complexity of the calibration assessment. Finally, transparent reporting cannot remedy an evaluation dataset that is poorly aligned with the intended target population.

The article is methodological and does not present a new empirical dataset. Its value should therefore be judged by the clarity, applicability, and verifiability of the proposed assessment sequence. Future work could convert the framework into outcome-specific reporting templates and evaluate whether structured result records reduce direction, rounding, and version errors across analysis, figures, and manuscripts.

6 Conclusion

Calibration slopes can be interpreted reliably only when the original prediction rule, target setting, analysis set, recalibration model, uncertainty, and graphical evidence are all explicit. A slope below one generally indicates predictions that are too extreme, whereas a slope above one indicates predictions that are not dispersed enough. These findings do not establish clinical usefulness or identify a unique updating strategy. Linking fitted objects to machine-readable results and then to tables, figures, and manuscript statements provides a practical basis for reproducible external model evaluation.

Declarations

Ethics and data: This methodological example does not analyze participant-level data and therefore does not require participant consent or ethics review. Funding, affiliations, and conflicts of interest are not supplied

because the demonstration has no named authors or institutions; they should not be fabricated to imitate a submitted article. Data availability: No empirical dataset was generated for this methods article. The document-generation source is available with the demonstration repository.

References

- [1] Steyerberg EW, Vickers AJ, Cook NR, et al. Assessing the performance of prediction models: a framework for traditional and novel measures. *Epidemiology*. 2010;21:128–138. doi:10.1097/EDE.0b013e3181c30fb2.
- [2] Van Calster B, McLernon DJ, van Smeden M, Wynants L, Steyerberg EW. Calibration: the Achilles heel of predictive analytics. *BMC Medicine*. 2019;17:230. doi:10.1186/s12916-019-1466-7.
- [3] Van Calster B, Nieboer D, Vergouwe Y, De Cock B, Pencina MJ, Steyerberg EW. A calibration hierarchy for risk models was defined: from utopia to empirical data. *Journal of Clinical Epidemiology*. 2016;74:167–176. doi:10.1016/j.jclinepi.2015.12.005.
- [4] Van Calster B, Vickers AJ. Calibration of risk prediction models: impact on decision-analytic performance. *Medical Decision Making*. 2015;35:162–169. doi:10.1177/0272989X14547233.